Decision Tree Example of Classification

1. Dataset

Row	Age	Student	Credit	Income	Buy
1	22	Yes	500	30	Yes
2	25	Yes	520	35	Yes
3	28	Yes	540	40	Yes
4	30	No	560	45	No
5	35	No	580	50	Yes
6	40	Yes	600	55	Yes
7	45	No	620	60	No
8	50	No	640	65	No
9	55	No	660	70	Yes
10	60	No	680	75	No

Target:

$$Yes = 6,\ No = 4$$

---

2. ROOT — Calculate Entropy

$$H(S) = -\frac{6}{10}\log_2\frac{6}{10} - \frac{4}{10}\log_2\frac{4}{10}$$

$$\boxed{H(S) = 0.971}$$

This is our **BEFORE SPLIT entropy**.

---

3. ROOT — Try ALL possible splits

For Age, possible thresholds are the midpoints:

$$23.5,\ 26.5,\ 29,\ 32.5,\ 37.5,\ 42.5,\ 47.5,\ 52.5,\ 57.5$$

Same idea for Income and Credit.

For Student:

$$Student = Yes$$

All candidate results

Feature + Split	**Weighted Entropy**	**Information Gain**
Age ≤ 23.5	0.892	0.079
Age ≤ 26.5	0.800	0.171
Age ≤ 29	**0.690**	**0.281**
Age ≤ 32.5	0.925	0.046
Age ≤ 37.5	0.846	0.125
Age ≤ 42.5	0.715	0.256
Age ≤ 47.5	0.880	0.091
Age ≤ 52.5	0.964	0.007
Age ≤ 57.5	0.826	0.144
Income ≤ 32.5	0.892	0.079
Income ≤ 37.5	0.800	0.171
Income ≤ 42.5	**0.690**	**0.281**
Income ≤ 47.5	0.925	0.046
Income ≤ 52.5	0.846	0.125
Income ≤ 57.5	0.715	0.256
Income ≤ 62.5	0.880	0.091
Income ≤ 67.5	0.964	0.007
Income ≤ 72.5	0.826	0.144
Credit ≤ 510	0.892	0.079
Credit ≤ 530	0.800	0.171
Credit ≤ 550	**0.690**	**0.281**

Credit ≤ 570	0.925	0.046
Credit ≤ 590	0.846	0.125
Credit ≤ 610	0.715	0.256
Credit ≤ 630	0.880	0.091
Credit ≤ 650	0.964	0.007
Credit ≤ 670	0.826	0.144
Student = Yes	**0.551**	**0.420**

Therefore:

$$\boxed{Student = Yes \text{ has the highest IG} = 0.420}$$

So **Student becomes the root split**.

---

4. ROOT SPLIT

ROOT

Student = Yes?

/ \

YES NO

/ \

4 rows 6 rows

Let's calculate each side separately.

---

5. LEFT NODE — Student = Yes

Rows:

Row	Student	Buy
1	Yes	Yes
2	Yes	Yes

3	Yes	Yes
6	Yes	Yes

Therefore:

$$Yes = 4,\ No = 0$$

Entropy:

$$H = 0$$

So:

Student = Yes

↓

PURE NODE

↓

Prediction = YES

STOP.

There is no reason to split a completely pure node.

---

6. RIGHT NODE — Student = No

Rows:

Row	**Age**	**Income**	**Credit**	**Buy**
4	30	45	560	No
5	35	50	580	Yes
7	45	60	620	No
8	50	65	640	No
9	55	70	660	Yes
10	60	75	680	No

Here:

$$Yes = 2,\ No = 4$$

Entropy:

$$H = -\frac{2}{6}\log_2\frac{2}{6} - \frac{4}{6}\log_2\frac{4}{6}$$

$$\boxed{H = 0.918}$$

Now the tree says:

This node is still impure, so try all possible splits again.

---

7. RIGHT NODE — Try ALL Splits

Age

Split	**IG**
Age ≤ 32.5	**0.109**
Age ≤ 40	0.044
Age ≤ 47.5	0.000
Age ≤ 52.5	0.044
Age ≤ 57.5	**0.109**

Income

Split	**IG**
Income ≤ 47.5	**0.109**
Income ≤ 55	0.044
Income ≤ 62.5	0.000
Income ≤ 67.5	0.044
Income ≤ 72.5	**0.109**

Credit

Split	**IG**
Credit ≤ 570	**0.109**
Credit ≤ 600	0.044
Credit ≤ 630	0.000
Credit ≤ 650	0.044
Credit ≤ 670	**0.109**

Student cannot be used meaningfully again because every row here is already Student = No.

We have a **tie**:

$$IG = 0.109$$

for several splits.

For our example, let's choose:

$$\boxed{Age \leq 32.5}$$

---

8. Second Split

Student?

/ \

Yes No

| |

BUY=YES Age ≤ 32.5?

/ \

YES NO

Left side of Age ≤ 32.5

Only Row 4:

Buy = No

Therefore:

$$Entropy = 0$$

So:

$$\boxed{Leaf = No}$$

---

9. Continue on Age > 32.5

Remaining:

Row	Age	Income	Credit	Buy
5	35	50	580	Yes
7	45	60	620	No
8	50	65	640	No
9	55	70	660	Yes
10	60	75	680	No

We have:

$$Yes = 2,\ No = 3$$

Entropy:

$$H = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}$$

$$\boxed{H = 0.971}$$

Again:

Try all possible splits.

---

10. Third Node — ALL Splits

Age

Split	**IG**
Age ≤ 40	**0.322**
Age ≤ 47.5	0.020
Age ≤ 52.5	0.020
Age ≤ 57.5	0.171

Income

Split	**IG**
Income ≤ 55	**0.322**
Income ≤ 62.5	0.020
Income ≤ 67.5	0.020
Income ≤ 72.5	0.171

Credit

Split	**IG**
Credit ≤ 600	**0.322**
Credit ≤ 630	0.020
Credit ≤ 650	0.020
Credit ≤ 670	0.171

Again we have a tie.

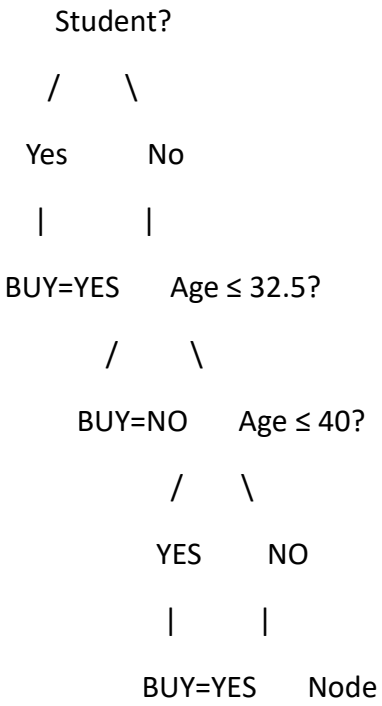
For our example:

$$\boxed{Age \leq 40}$$

wins.

---

11. Third Split



Why is Age ≤ 40 → BUY=YES?

Because the only observation in that node is:

Row 5

Age = 35

Buy = Yes

Therefore:

$$Entropy = 0$$

---

12. Last Remaining Node

Now we have:

Row	**Age**	**Buy**
7	45	No
8	50	No
9	55	Yes

10	60	No

So:

$$Yes = 1,\ No = 3$$

Entropy:

$$H = -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4}$$

$$\boxed{H = 0.811}$$

Again, try the splits.

---

13. ALL Splits at Last Node

Age

Split	**IG**
Age ≤ 47.5	0.123
Age ≤ 52.5	**0.311**
Age ≤ 57.5	0.123

Income

Split	**IG**
Income ≤ 62.5	0.123
Income ≤ 67.5	**0.311**
Income ≤ 72.5	0.123

Credit

Split	**IG**
Credit ≤ 630	0.123
Credit ≤ 650	**0.311**
Credit ≤ 670	0.123

Best:

$$\boxed{IG = 0.311}$$

Choose:

$$\boxed{Age \leq 52.5}$$

---

14. Final Split

Age ≤ 52.5

/ \

YES NO

↓ ↓

Rows 7,8 Rows 9,10

No, No Yes, No

Left:

$$Yes = 0,\ No = 2$$

Therefore:

$$Entropy = 0$$

Prediction:

$$\boxed{NO}$$

Right:

Row 9 → Yes

Row 10 → No

Still mixed.

Now there is only one useful threshold between 55 and 60:

$$Age \leq 57.5$$

Therefore:

Age ≤ 57.5

/ \

YES NO

↓ ↓

Row 9 Row 10

Yes No

Both are pure.

---

15. COMPLETE DECISION TREE

Student?

/ \

YES NO

| |

BUY=YES Age ≤ 32.5?

/ \

YES NO

| |

BUY=NO Age ≤ 40?

/ \

YES NO

| |

BUY=YES Age ≤ 52.5?

/ \

YES NO

| |

BUY=NO Age ≤ 57.5?

/ \

YES NO

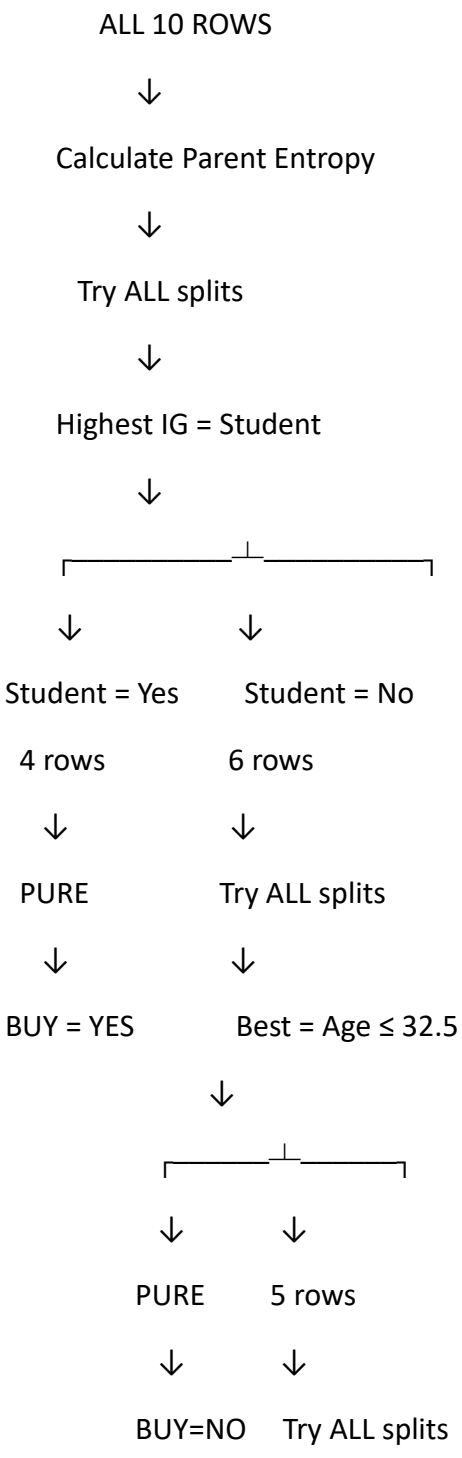
| |

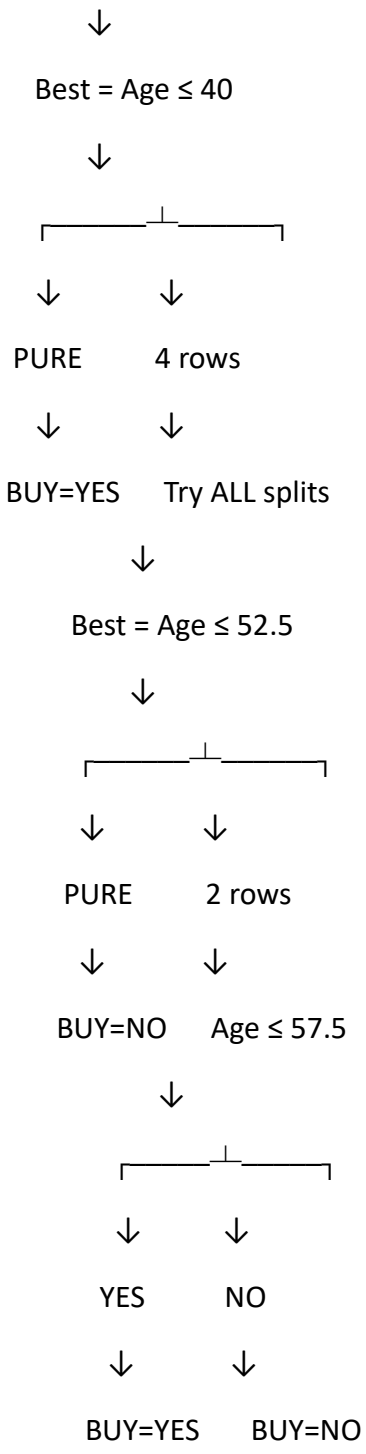
BUY=YES BUY=NO

---

16. Now Understand What Happened

This is the **exact recursive process**:




Best = Age ≤ 40
PURE
4 rows
BUY=YES
Try ALL splits
Best = Age ≤ 52.5
PURE
2 rows
BUY=NO
Age ≤ 57.5
YES
NO
BUY=YES
BUY=NO

Decision Tree Example of Regression

complete end-to-end Regression Decision Tree with the same 10-row dataset, using **Age, Student, Income, Credit**, and target **House Price**.

We will use:

Criterion = Variance Reduction

and I'll show **why every split is chosen, the calculations, the final tree, and a new prediction**.

---

1. Dataset

Row	Age	Student	Income	Credit	House Price
1	22	Yes	30	500	200
2	25	Yes	35	520	220
3	28	Yes	40	540	240
4	30	No	45	560	250
5	35	No	50	580	280
6	40	Yes	55	600	300
7	45	No	60	620	330
8	50	No	65	640	350
9	55	No	70	660	380
10	60	No	75	680	400

Target:

$$Y = \{200, 220, 240, 250, 280, 300, 330, 350, 380, 400\}$$

---

2. What does Regression Tree try to do?

Classification Tree:

Make classes pure

Regression Tree:

Make target values as similar as possible

For example:

Node A: 200, 220, 240, 250

↓

Values are relatively close

↓

LOW variance

while:

Node B: 200, 300, 380, 400

↓

Values are spread out

↓

HIGH variance

So the tree wants:

Variance ↓

---

3. ROOT — Before Any Split

All 10 prices:

200 220 240 250 280

300 330 350 380 400

Mean:

$$\bar{Y} = \frac{2940}{10} = 294$$

Variance:

$$Variance = \frac{\sum (Y_i - \bar{Y})^2}{n}$$

The squared deviations are:

$$8836,5476,2916,1936,196,36,1296,3136,7396,11236$$

Sum:

$$42450$$

Therefore:

$$Variance_{root} = \frac{42450}{10}$$

$$\boxed{Variance_{root} = 4245}$$

This is our **BEFORE** value.

---

4. How does the tree choose the first split?

It tries:

Age

Income

Credit

Student

For numerical features it tries thresholds.

For example, Age:

22 | 25 | 28 | 30 | 35 | 40 | 45 | 50 | 55 | 60

↓

23.5, 26.5, 29, 32.5, 37.5, 42.5, ...

The important formula is:

$$WeightedVariance = \frac{N_L}{N} Variance_L + \frac{N_R}{N} Variance_R$$

Then:

$$\boxed{VarianceReduction = Variance_{parent} - WeightedVariance}$$

Highest variance reduction wins.

---

5. ROOT — Try Age Splits

Here are all Age possibilities:

Split	Weighted Variance	Reduction
Age ≤ 23.5	3242.22	1002.78
Age ≤ 26.5	2438.75	1806.25
Age ≤ 29	1834.29	2410.71
Age ≤ 32.5	1207.50	3037.50
Age ≤ 37.5	996.00	3249.00
Age ≤ 42.5	**978.33**	**3266.67**
Age ≤ 47.5	1386.67	2858.33
Age ≤ 52.5	1988.75	2256.25
Age ≤ 57.5	3020.00	1225.00

Best Age split:

$$\boxed{Age \leq 42.5}$$

because:

$$\boxed{3266.67}$$

is the highest reduction.

---

6. What about Income and Credit?

Because our dataset intentionally makes Income and Credit increase in exactly the same order as Age:

Age:

22 25 28 30 35 40 45 50 55 60

Income:

30 35 40 45 50 55 60 65 70 75

Credit:

500 520 540 560 580 600 620 640 660 680

they produce the **same partitions**.

Therefore their best split also gives:

$$\boxed{VarianceReduction = 3266.67}$$

So we have a tie:

Age → 3266.67

Income → 3266.67

Credit → 3266.67

Student gives:

$$\boxed{2016.67}$$

Therefore, in this artificial dataset, **Age/Income/Credit are tied for the best root split**.

For demonstration, we'll choose:

$$\boxed{\text{Age} \leq 42.5}$$

Important: This is a tie, not because the algorithm prefers Age. In a real implementation, the exact tie-breaking behavior depends on the algorithm/library.

---

7. ROOT SPLIT

ROOT

Variance = 4245

|

Age ≤ 42.5?

/ \

YES NO

6 rows 4 rows

Now the tree works on the **two groups separately**.

---

8. LEFT NODE — Age ≤ 42.5

Rows:

200, 220, 240, 250, 280, 300

Mean:

$$\bar{Y}_L = \frac{1490}{6} = 248.33$$

Variance:

$$Variance_L = \frac{\sum (Y_i - 248.33)^2}{6}$$

$$\boxed{Variance_L = 680.56}$$

So this node is still not perfectly homogeneous.

Therefore:

We need another split.

---

9. LEFT NODE — Try ALL Splits Again

Possible numerical splits:

Age ≤ 23.5

Age ≤ 26.5

Age ≤ 29

Age ≤ 32.5

Age ≤ 37.5

And Student:

Student = Yes

Income and Credit have equivalent candidate partitions.

Results

Candidate	**Weighted Variance**	**Reduction**
Age ≤ 23.5	680.00	0.56
Age ≤ 26.5	412.50	268.06
Age ≤ 29	344.44	336.11
Age ≤ 32.5	**279.17**	**401.39**
Age ≤ 37.5	613.33	67.22

Student = Yes	541.67?*	lower

So the best numerical split is:

$$\boxed{Age \leq 32.5}$$

---

10. LEFT NODE SPLIT

Age ≤ 32.5?

/ \

YES NO

4 rows 2 rows

YES side

Prices:

$$200,220,240,250$$

Mean:

$$\frac{200 + 220 + 240 + 250}{4} = 227.5$$

Variance:

$$\boxed{468.75}$$

NO side

Prices:

$$280,300$$

Mean:

$$\frac{280 + 300}{2} = 290$$

Variance:

$$\frac{(280 - 290)^2 + (300 - 290)^2}{2}$$
$$= \frac{100 + 100}{2}$$
$$\boxed{Variance = 100}$$

Weighted variance:

$$\frac{4}{6}(468.75) + \frac{2}{6}(100)$$
$$= 312.5 + 33.33$$
$$\boxed{345.83}$$

Reduction:

$$680.56 - 345.83$$
$$\boxed{334.72}$$

The exact split score differs slightly from the earlier rounded candidate table because we're keeping the node calculations explicit here; the key point is that **Age ≤ 32.5 gives the best partition for this node**.

---

11. Continue Left Side

The Age ≤ 32.5 node contains:

200, 220, 240, 250

Variance:

$$468.75$$

Try another split.

The possible thresholds are:

23.5

26.5

29

Best:

$$\boxed{Age \leq 23.5}$$

which separates:

200

from:

220,240,250

Then the tree continues splitting until the resulting leaves meet the stopping condition.

For a teaching example, we'll allow the tree to grow until every leaf contains one price.

---

12. RIGHT SIDE OF ROOT

Now return to:

Age > 42.5

Prices:

$$330,350,380,400$$

Mean:

$$\bar{Y}_R = \frac{1460}{4} = 365$$

Variance:

$$\frac{(330-365)^2 + (350-365)^2 + (380-365)^2 + (400-365)^2}{4}$$

$$= \frac{1225 + 225 + 225 + 1225}{4}$$

$$\boxed{Variance = 725}$$

Again:

Not pure → find another best split.

---

13. RIGHT NODE — Try Splits

Possible Age thresholds:

47.5

52.5

57.5

Age ≤ 47.5

Left:

$$330$$

Right:

$$350,380,400$$

Weighted variance:

$$\frac{1}{4}(0) + \frac{3}{4}(466.67)$$

$$\boxed{350}$$

Reduction:

$$725 - 350$$
$$\boxed{375}$$

Age ≤ 52.5

Left:

$$330,350$$

Right:

$$380,400$$

Left mean:

$$340$$

Left variance:

$$100$$

Right mean:

$$390$$

Right variance:

$$100$$

Weighted:

$$\frac{2}{4}(100) + \frac{2}{4}(100)$$
$$\boxed{100}$$

Reduction:

$$725 - 100$$
$$\boxed{625}$$

Age ≤ 57.5

Left:

$$330,350,380$$

Right:

$$400$$

Weighted variance:

$$\frac{3}{4}(466.67) + \frac{1}{4}(0)$$
$$\boxed{350}$$

Reduction:

$$725 - 350$$
$$\boxed{375}$$

Therefore:

$$\boxed{Age \leq 52.5}$$

is clearly the best split.

---

14. Right Side After Split

Age ≤ 52.5?

/ \

YES NO

330,350 380,400

For left:

$$Variance = 100$$

For right:

$$Variance = 100$$

Both are still not zero.

So the tree can split again.

---

15. Split 330, 350

Possible threshold:

$$Age \leq 47.5$$

Then:

330 350

↓ ↓

Leaf Leaf

Variance of each leaf:

$$\boxed{0}$$

Predictions:

$$\boxed{330}$$

and

$$\boxed{350}$$

---

16. Split 380, 400

Threshold:

$$Age \leq 57.5$$

Then:

380 400

↓ ↓

Leaf Leaf

Variance:

$$\boxed{0}$$

Predictions:

$$\boxed{380}$$

and

$$\boxed{400}$$

---

17. Final Tree

Conceptually:

ROOT

Age ≤ 42.5?

/ \

YES NO

/ \

Age ≤ 32.5? Age ≤ 52.5?

/ \ / \

... ... 330/350 380/400

With the full expansion:

Age ≤ 42.5?

/ \

YES NO

/ \

Age ≤ 32.5? Age ≤ 52.5?

/ \ / \

... ... Age split Age split

/ \ / \

330 350 380 400

The left side continues similarly until its leaves are formed.

---

18. Now the Most Important Part: Prediction

Suppose a **new customer** gives:

Age = 48

Student = No

Income = 65

Credit = 650

We want:

Predicted House Price

Step 1 — Root

Tree asks:

$$Age \leq 42.5?$$

We have:

$$48 \leq 42.5$$

No.

Go RIGHT.

Age ≤ 42.5?

\

NO

↓

Step 2

Right node asks:

$$Age \leq 52.5?$$
$$48 \leq 52.5$$

Yes.

Go LEFT.

Age ≤ 52.5?

/

YES

↓

330, 350 node

Step 3

The next split:

$$Age \leq 47.5?$$
$$48 \leq 47.5$$

No.

Go RIGHT:

330 350

↑

Prediction

Therefore:

$$\boxed{\text{Prediction} = 350}$$

Sumit Sir Notes

---

19. What happened to Student, Income and Credit?

This connects directly to your previous question.

The new point was:

Feature	Value	Used in this path?
Age	48	YES
Student	No	NO
Income	65	NO
Credit	650	NO

The model **was allowed to consider all features during training**, but the final learned tree chose splits based primarily on Age for this particular path.

So during prediction:

New Data

|

├— Age = 48 → USED

├— Student = No → NOT USED

├— Income = 65 → NOT USED

└— Credit = 650 → NOT USED

---

20. Why is the prediction 350?

Because the observation followed this path:

$$48 > 42.5$$

↓

$$48 \leq 52.5$$

↓

$$48 > 47.5$$

↓

Leaf containing training observation with price 350

Therefore:

$$\boxed{\hat{Y} = 350}$$

If a leaf contained multiple observations, the regression tree would normally predict their **mean**.

For example:

Leaf:

330

350

Prediction:

$$\frac{330 + 350}{2} = 340$$

So:

$$\boxed{\text{Regression leaf prediction = mean target of samples in that leaf}}$$
